Future Founders: Data-Centric Agents

From Prompt to Production Data. Build a product where AI agents reason over, act on and learn from real data. Solve a clear customer problem in one day, with agents grounded in live, changing data rather than a static prompt.

The Vision

Most agent demos run on a prompt and a few tool calls. Real products need agents that work with data: data that changes, streams in, goes stale, conflicts with other data or builds up over time. The best agents will know what data they're using, keep it fresh, remember what they learned and show their working.

We want to see products where the data layer is what makes the agent useful, not an afterthought.

Your Challenge

Build a working product where one or more agents depend on data to deliver value for a specific user. That could mean querying large datasets, reacting to event streams, keeping long-term memory, combining many sources or improving over time from past runs. You choose the customer and the problem.

Use any models, frameworks and tools you like. Deploy your app to our sponsor Aiven's Runtime, and use their managed data infrastructure for your data layer where relevant (PostgreSQL, ClickHouse, Valkey, OpenSearch and more) with free credits for every team. Strong teams need more than code. Product thinking, design and a credible route to first users all count.

Need Inspiration?

1. The Analyst Agent

An agent that answers business questions over a large, messy dataset. It writes and checks its own queries, says how confident it is, and cites the exact rows behind each answer.

2. The Event-Driven Agent

Agents that subscribe to a live stream (payments, IoT sensors, support tickets, logs) and decide, escalate or act in real time rather than waiting for a prompt.

3. The Agent With a Memory

An assistant that builds durable memory of a customer, codebase or domain across sessions. Show that it gets measurably better over time, and how it handles memories that are stale or contradict each other.

4. The Data-to-Answer Detective

Trace a wrong agent answer back to its source. Was it a stale record, a failed tool call or an unsupported claim? Then show that fixing the data fixes the answer.

5. (Advanced) The Self-Improving Agent

Store traces, evaluations and configurations, then let an optimiser agent mine past runs to propose changes to retrieval, context or tool use. Keep the scope tight: pick one improvement target and show a measured before-and-after comparison.

What to Show in Your Demo

- A working product that solves the customer problem in one end-to-end user journey
- The data layer in action: what data the agent uses, where it lives and how it flows
- Evidence it works: a before-and-after, a benchmark, or a comparison against a naive or no-data baseline
- One pitch slide: who it's for, why now, and how you'd reach your first users
- Use data that's safe to share publicly. Make clear which results come from live runs and which come from recorded scenarios.

Resources & Support

- **Aiven credits:** every team gets cloud credits for Aiven's managed data services. To claim them, create a new account on Aiven
- **Docs & templates:** aiven.io/docs and templates.aiven.io
- **On-site mentors:** experts from the organising team and our sponsors can help with technical questions, product questions or anything else you need.

Judging Criteria

1. **Customer Problem & Product Value:** Is it a real problem for a clear user, and would they adopt the product?
2. **Agent Capability:** Does it show advanced agent behaviour, such as multi-step reasoning, autonomy, teamwork or learning, beyond a single LLM call?
3. **Data at the Core:** Does the product depend on using data well (freshness, scale, grounding, memory), or is the data just decoration?

4. **Evidence It Works:** Is there a measurable result, such as accuracy, cost, latency, success rate or time saved?
5. **Execution & Pitch:** Is it a polished, understandable demo with a clear point of difference and a credible route to first users?

Sponsor prize: Aiven will award a separate prize for the best use of Aiven's data platform.